# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

For LED's to emit light in visible region of electromagnetic light, it should have energy band gap in the range of:

**(A)** $0.1\text{eV}$ to $0.4\text{eV}$

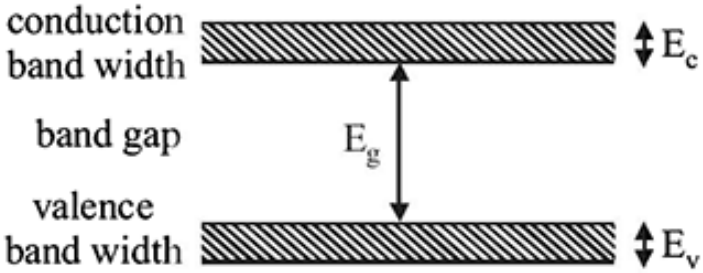
**(B)** $0.5\text{eV}$ to $0.8\text{eV}$

**(C)** $0.9\text{eV}$ to $1.6\text{eV}$

**(D)** $1.7\text{eV}$ to $3.0\text{eV}$

**Q.2**

If the lattice constant of this semiconductor is decreased, then which of the following is correct?




**(A)** All $E_c, E_g, E_v$ increase

**(B)** $E_c$ and $E_v$ increase, but $E_g$ decreases

**(C)** $E_c$ and $E_v$ decrease, but $E_g$ increases

**(D)** All $E_c, E_g, E_v$ decrease

**Q.3**

According to Band Theory of Solids, what can be said about the relationship between band-gap width and the tendency of material to conduct electricity?

**(A)** The lesser the band-gap width the more insulating is the material.

**(B)** The greater the band-gap width the more conductive is the material.

**(C)** The greater the band-gap width the less conductive is the material.

**(D)** None of the above.

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**Q.4**

For LED's to emit light in visible region of electromagnetic light, it should have energy band gap in the range of:

**(A)** $0.1\text{eV}$ to $0.4\text{eV}$

**(B)** $0.5\text{eV}$ to $0.8\text{eV}$

**(C)** $0.9\text{eV}$ to $1.6\text{eV}$

**(D)** $1.7\text{eV}$ to $3.0\text{eV}$

**Q.5**

The ratio of electron and hole currents in a semiconductor is $7/4$ and the ratio of drift velocities of electrons and holes is $5/4$, then ratio of concentrations of electrons and holes will be

**(A)** $5/7$

**(B)** $7/5$

**(C)** $25/49$

**(D)** $49/25$

**Q.6**

In germanium the energy gap is about $0.75\text{eV}$. The wavelength of light which germanium starts absorbing is

**(A)** $5000\mathring{A}$

**(B)** $1650\mathring{A}$

**(C)** $16500\mathring{A}$

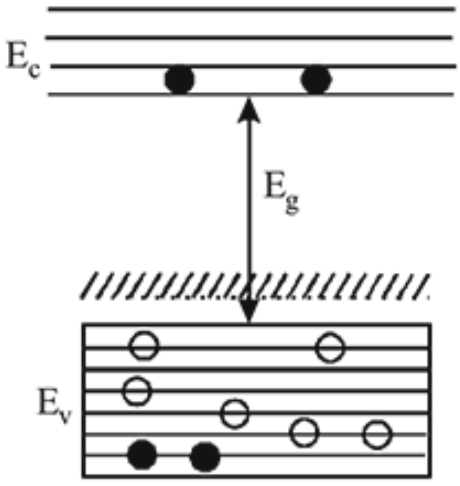
**(D)** $165000\mathring{A}$

**Q.7**

In the energy band diagram of a material $\text{E}_\text{c}$ shown below, the open circles and filled circles denote holes

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and electrons respectively. The material is




**(A)** an insulator

**(B)** a metal

**(C)** an $\mathrm{n}$ -type semiconductor

**(D)** a p-type semiconductor

**Q.8**

A pure semiconductor has equal electron and hole concentration of $10^{16}\ \mathrm{m}^{-3}$. Doping by indium increases number of hole concentration $n_h$ to $5 \times 10^{22}\ \mathrm{m}^{-3}$. Then, the value of number of electron concentration $n_e$ in the doped semiconductor is

**(A)** $10^6/\mathrm{m}^3$

**(B)** $10^{22}/\mathrm{m}^3$

**(C)** $2 \times 10^6/\mathrm{m}^3$

**(D)** $2 \times 10^9/\mathrm{m}^3$

**Q.9**

The concentration of hole-electron pairs in pure silicon at $\mathrm{T} = 300\ \mathrm{K}$ is $7 \times 10^{15}$ per cubic meter. Antimony is doped into silicon in a proportion of 1 atom in $10^7$ Si atoms. Assuming half of the impurity atoms contribute electron in the conduction band, calculate the factor by which the number of charge carriers increases due to doping. The number of silicon atoms per cubic meter is $5 \times 10^{28}$

**(A)** $2.8 \times 10^5$

**(B)** $3.1 \times 10^2$

**(C)** $4.2 \times 10^5$

**(D)** $1.8 \times 10^5$

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**Q.10**

In $p$-$n$ junction diode, holes diffuse from $p$-region to $n$-region because

**(A)** the free electrons in the $n$-region attract them.

**(B)** there is a greater concentration of holes in $p$-region as compared to $n$-region.

**(C)** they are swept across the junction by potential difference.

**(D)** there is great concentration of electrons in $p$-region as compared to $n$-region.

**Q.11**

The intrinsic conductivity of germanium at $27^{\circ}$ is $2.13 \mathrm{mho\ m^{-1}}$ and mobilities of electrons and holes are $0.38$ and $0.18\ \mathrm{m^2\ V^{-1}\ s^{-1}}$ respectively. The density $\left(\mathrm{inm^{-3}}\right)$ of charge carriers is Assuming 2 decimal places in answer expression, what is the power of $10$?

**(A)** 15

**(B)** 17

**(C)** 14

**(D)** 19

**Q.12**

The electric conductivity of an intrinsic semiconductor increases when the electromagnetic waves of wavelength equal or shorter than $3895\ \mathrm{nm}$ is incident on it. The band gap of the semiconductor is

**(A)** $0.2\mathrm{eV}$

**(B)** $0.3\mathrm{eV}$

**(C)** $1.0\mathrm{eV}$

**(D)** $1.5\mathrm{eV}$

**Q.13**

The concentration of hole - electron pairs in pure silicon at $\mathrm{T} = 300\ \mathrm{K}$ is $7 \times 10^{15}$ per cubic meter. Antimony is doped into silicon in a proportion of 1 atom in $10^7$ Si atoms. Assuming that half of the impurity atoms contribute electron in the conduction band, calculate the factor by which the number of charge carriers increases due to doping. The number of silicon atoms per cubic meter is $5 \times 10^{28}$

**(A)** $2.8 \times 10^5$

**(B)** $3.1 \times 10^2$

**(C)** $4.2 \times 10^5$

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**(D)** $1.8 \times 10^5$

**Q.14**

If the ratio of the concentration of electrons to that of holes in a semiconductor is $\frac{7}{5}$ and the ratio of currents is $\frac{7}{4}$, then what is the ratio of their drift velocities?

**(A)** $\frac{5}{8}$

**(B)** $\frac{4}{5}$

**(C)** $\frac{5}{4}$

**(D)** $\frac{4}{7}$

**Q.15**

Pure Si at $500\ \mathrm{K}$ has equal number of electron $\left(\mathrm{n_e}\right)$ and hole $\left(\mathrm{n_h}\right)$ concentrations of $1.5 \times 10^{16}\ \mathrm{m}^{-3}$. Doping by indium increases $n_h$ to $4.5 \times 10^{22}\ \mathrm{m}^{-3}$. The doped semiconductor is of

**(A)** n-type with electron concentration $\mathrm{n_e} = 5 \times 10^{22}\ \mathrm{m}^{-3}$

**(B)** p-type with electron concentration $\mathrm{n_e} = 2.5 \times 10^{10}\ \mathrm{m}^{-3}$

**(C)** $\mathrm{n}$ -type with electron concentration $\mathrm{n_e} = 2.5 \times 10^{23}\ \mathrm{m}^{-3}$

**(D)** p-type having electron concentration $\mathrm{n_e} = 5 \times 10^{9}\ \mathrm{m}^{-3}$

**Q.16**

In a p-type semiconductor the acceptor level is situated $60\mathrm{meV}$ above the valence band. The maximum wavelength of light required to produce hole will be

**(A)** $0.207 \times 10^{-5}\ \mathrm{m}$

**(B)** $2.07 \times 10^{-5}\ \mathrm{m}$

**(C)** $20.7 \times 10^{-5}\ \mathrm{m}$

**(D)** $2075 \times 10^{-5}\ \mathrm{m}$

**Q.17**

Carbon, silicon and germanium have four valence electrons each. At room temperature which one of the following statements is most appropriate?

**(A)** The number of free electrons for conduction is significant only in $\mathrm{Si}$ and Ge but small in $\mathrm{C}$.

**(B)** The number of free conduction electrons is significant in $\mathrm{C}$ but small in $\mathrm{Si}$ and Ge.

**(C)** The number of free conduction electrons is negligibly small in all the three.

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**(D)** The number of free electrons for conduction is significant in all the three.

**Q.18**

On doping germanium with donor atoms of density $10^{17}\ \mathrm{cm}^{-3}$ its conductivity in $\mathrm{mho/cm}$ will be [Given $\mu_e = 3800\ \mathrm{cm}^2/\mathrm{V-s}$ and $\mathrm{n_i} = 2.5 \times 10^{13}\ \mathrm{cm}^{-13}$]

**(A)** $30.4$

**(B)** $60.8$

**(C)** $91.2$

**(D)** $121.6$

**Q.19**

The intrinsic conductivity of germanium at $27^\circ$ is $2.13\mathrm{mhom}^{-1}$ and mobilities of electrons and holes are $0.38$ and $0.18\ \mathrm{m}^2\ \mathrm{V}^{-1}\ \mathrm{s}^{-1}$ respectively. The density of charge carriers is

**(A)** $2.37 \times 10^{19}\ \mathrm{m}^{-3}$

**(B)** $3.28 \times 10^{19}\ \mathrm{m}^{-3}$

**(C)** $7.83 \times 10^{19}\ \mathrm{m}^{-3}$

**(D)** $8.47 \times 10^{19}\ \mathrm{m}^{-3}$

**Q.20**

By increasing the temperature, the specific resistance of a conductor and a semiconductor

**(A)** increases for both

**(B)** decreases for both

**(C)** increases, decreases

**(D)** decreases, increases

## Answers & Solutions

**Q.1 Answer:**

$1.7\mathrm{eV}$ to $3.0\mathrm{eV}$

**Solution:**

**Energy band gap range is given by,** $\mathrm{E_g} = \frac{\mathrm{hc}}{\lambda}$

**For visible region** $\lambda = \left(4 \times 10^{-7} \sim 7 \times 10^{-7}\right)\ \mathrm{m}$

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$$E_g = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{7 \times 10^{-7}}$$
$$= \frac{19.8 \times 10^{-26}}{7 \times 10^{-7}}$$
$$= \frac{2.8 \times 10^{-19}}{1.6 \times 10^{-19}}$$
$$E_g = 1.75 \text{eV}$$

**Q.2 Answer:**

$E_c$ and $E_v$ decrease, but $E_g$ increases

**Solution:**

**A crystal structure is composed of a unit cell, a set of atoms arranged in a particular way; which is periodically repeated in three dimensions on a lattice. The spacing between unit cells in various directions is called its lattice parameters or constants. Increasing these lattice constants will increase or widen the band-gap $(E_g)$, which means more energy would be required by electrons to reach the conduction band from the valence band. Automatically $E_c$ and $E_v$ decreases.**

**Q.3 Answer:**

The greater the band-gap width the less conductive is the material.

**Solution:**

**Band-gap is the distance between the valence band of electrons and the conduction band. According to Band Theory of Solids, the greater the band-gap width the less conductive is the material.**

**Q.4 Answer:**

$1.7 \text{eV}$ to $3.0 \text{eV}$

**Solution:**

**No solution available**

**Q.5 Answer:**

$7/5$

**Solution:**

$$I = nAev_d \text{ or } I \propto nv_d$$
$$\frac{I_e}{I_h} = \frac{n_e v_e}{n_h v_h} \text{ or } \frac{n_e}{n_h} = \frac{I_e}{I_h} \times \frac{v_h}{v_e} = \frac{7}{4} \times \frac{4}{5} = \frac{7}{5}$$

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**Q.6 Answer:**

$16500\mathring{A}$

**Solution:**

$\lambda = \frac{hc}{E} = \frac{6.63\times10^{-34}\times3\times10^{8}}{0.75\times1.6\times10^{-19}} = 16500\mathring{A}$

**Q.7 Answer:**

a p-type semiconductor

**Solution:**

**No solution available**

**Q.8 Answer:**

$2 \times 10^{9}/\mathrm{m}^{3}$

**Solution:**

$$\text{Here, } \mathrm{n_i} = 10^{16}\ \mathrm{m^{-3}}, \mathrm{n_h} = 5\times10^{22}\ \mathrm{m^{-3}}$$
$$\text{As } \mathrm{n_e n_h} = \mathrm{n_i^2}$$
$$\therefore \mathrm{n_e} = \frac{\mathrm{n_i^2}}{\mathrm{n_h}} = \frac{\left(10^{16}\ \mathrm{m^{-3}}\right)^2}{5\times10^{22}\ \mathrm{m^{-3}}} = 2\times10^{9}\ \mathrm{m^{-3}}$$

**Q.9 Answer:**

$1.8 \times 10^{5}$

**Solution:**

**In pure semiconductor electron-hole pair** $= 7 \times 10^{15}/\mathrm{m}^{3}$ $\mathrm{n_{initial}} = \mathrm{n_h} + \mathrm{n_e} = 14\times10^{15}$
**after doping donor Impurity**
$\mathrm{N_D} = \frac{5\times10^{28}}{10^{7}} = 5\times10^{21}$ **and** $\mathrm{n_e} = \frac{\mathrm{N_D}}{2} = 2.5\times10^{21}$
**So,** $n_{\text{final}} = n_h + n_e$
$\Rightarrow \mathrm{n_{final}} \approx n_e \approx 2.5\times10^{21}\ (\because \mathrm{n_e} >> \mathrm{n_h})$
**Factor** $= \frac{\mathrm{n_{final}} - \mathrm{n_{initial}}}{\mathrm{n_{initial}}}$
$= \frac{2.5\times10^{21} - 14\times10^{15}}{14\times10^{15}} \approx \frac{2.5\times10^{21}}{14\times10^{15}} = 1.8\times10^{5}$

**Q.10 Answer:**

there is a greater concentration of holes in $p$-region as compared to $n$-region.

**Solution:**

**In a $p$-$n$ junction diode, majority charge carriers diffuse from one side of the barrier to other side due to concentration gradient. As, the concentration of holes are greater in $p$-region as compared to $n$-region. Hence, diffusion of holes takes place from $p$-region to $n$-region.**

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**Q.11 Answer:**

19

**Solution:**

$\left(2.37 \times 10^{19}\right)$ **Conductivity,** $\sigma = \frac{1}{\rho} = \mathrm{e}\left(\mathrm{n_e}\mu_\mathrm{e} + \mathrm{n_h}\mu_\mathrm{h}\right)$

$2.13 = 1.6 \times 10^{-19}(0.38 + 0.18)\mathrm{n_i}$

**(Since in intrinsic semi-conductor,** $n_e = n_h = n_i$ **)** $\therefore$ **density of charge carriers,** $\mathrm{n_i}$

$= \frac{2.13}{1.6\times10^{-19}\times0.56} = 2.37 \times 10^{19}\ \mathrm{m}^{-3}$

**Q.12 Answer:**

$0.3\mathrm{eV}$

**Solution:**

**Forbidden energy gap** $= \frac{12400(\mathrm{eV}-\mathring{A})}{\lambda(\mathring{A})}$

$= \frac{12400}{38950}\mathrm{eV} = 0.3\mathrm{eV}$

**Q.13 Answer:**

$1.8 \times 10^5$

**Solution:**

**In pure semiconductor electron-hole pair** $= 7 \times 10^{15}/\mathrm{m}^3$

$\mathrm{n_{initial}} = \mathrm{n_h} + \mathrm{n_e} = 14 \times 10^{15}$ **after doping**

**donor Impurity** $\mathrm{N_D} = \frac{5\times10^{28}}{10^7} = 5 \times 10^{21}$ **and** $\mathrm{n_e} = \frac{\mathrm{N_D}}{2}$

$= 2.5 \times 10^{21}$

**So,** $\mathrm{n_{final}} = \mathrm{n_h} + \mathrm{n_e}$

$\Rightarrow \mathrm{n_{final}} \approx n_\mathrm{e} \approx 2.5 \times 10^{21}\ (\because \mathrm{n_e} \gg \mathrm{n_h})$

**Factor** $= \frac{\mathrm{n_{final}} - \mathrm{n_{initial}}}{\mathrm{n_{initial}}}$

$= \frac{2.5\times10^{21} - 14\times10^{15}}{14\times10^{15}} \approx \frac{2.5\times10^{21}}{14\times10^{15}} = 1.8 \times 10^5$

**Q.14 Answer:**

$\frac{5}{4}$

**Solution:**

$\frac{I_e}{I_h} = \frac{n_e e A v_e}{n_h e A v_h} \Rightarrow \frac{7}{4} = \frac{7}{5} \times \frac{v_e}{v_h} \Rightarrow \frac{v_e}{v_h} = \frac{5}{4}$

**Q.15 Answer:**

p-type having electron concentration $\mathrm{n_e} = 5 \times 10^9\ \mathrm{m}^{-3}$

**Solution:**

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$n_i^2 = n_e n_h$
$\left(1.5 \times 10^{16}\right)^2 = n_e \left(4.5 \times 10^{22}\right)$
$\Rightarrow n_e = 0.5 \times 10^{10}$ **or** $n_e = 5 \times 10^9$
**Given** $n_h = 4.5 \times 10^{22} \Rightarrow n_h >> n_e$
**$\therefore$ Semiconductor is p-type and** $n_e = 5 \times 10^9\ \mathrm{m}^{-3}$

**Q.16 Answer:**

$2.07 \times 10^{-5}\ \mathrm{m}$

**Solution:**

$$\lambda = \frac{hc}{E}$$
$$= \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{(60 \times 10^{-3} \times 1.6 \times 10^{-19})} = 2.07 \times 10^{-5}$$

**Q.17 Answer:**

The number of free electrons for conduction is significant only in Si and Ge but small in C.

**Solution:**

**No solution available**

**Q.18 Answer:**

60.8

**Solution:**

**Conductivity** $\sigma = n_i e \mu_e = 10^{17} \times \left(1.6 \times 10^{-19}\right) \times 3800$
$= 60.8 \mathrm{mho/cm}$

**Q.19 Answer:**

$2.37 \times 10^{19}\ \mathrm{m}^{-3}$

**Solution:**

**Conductivity,** $\sigma = \frac{1}{\rho} = e\left(n_e \mu_e + n_h \mu_h\right)$
**i.e.,** $2.13 = 1.6 \times 10^{-19}(0.38 + 0.18) n_i$
**(Since, in intrinsic semi-conductor,** $n_e = n_h = n_i$ **)**
**$\therefore$ density of charge carries,** $n_i = \frac{2.13}{1.6 \times 10^{-19} \times 0.56} = 2.37 \times 10^{19}/\mathrm{m}^3$

**Q.20 Answer:**

increases, decreases

**Solution:**

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**he resistivity of conductor increases with increase in temperature. The resistivity of semiconductor decreases as the temperature increases**

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